4 A light-dependent resistor (LDR) is connected across the terminals of a cell that has internal resistance. A voltmeter is connected across the LDR.

(a) Draw a circuit diagram to show this arrangement.

[3]

(b) The cell has an electromotive force (e.m.f.) of 4.90 V.

At a particular light intensity, the LDR has a resistance of 230 Ω and a potential difference (p.d.) across it of 4.78 V.

(i) Identify **one** possible source of systematic error in measuring the p.d. across the LDR using a digital voltmeter.

....

.... [1]

(ii) Calculate the power dissipated in the LDR.

power = ... W [2]

(iii) Determine the internal resistance of the cell.

internal resistance = ... Ω [2]

(iv) The intensity of the light illuminating the LDR is gradually decreased.

By placing **one** tick (✓) in each row, complete Table 4.1 to show what happens to the quantities indicated as the light intensity decreases.

Table 4.1

	decreases	**unchanged**	**increases**
resistance of LDR			
current in cell			
p.d. across LDR			

[3]

[Total: 11]